SRM

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Delhi-NCR Campus, Modinagar, Ghaziabad (U.P.)

ASSIGNMENT- IV

(B. Tech. III -Year, VI-SEMESTER)

SUBJECT NAME: DATA SCIENCE

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Submitted To:

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Q-1) Explain the Concept of Data Wrangling and its importance in Data Analysis.

→ Data Wrangling is the process of gathering, collecting, cleaning and transforming raw data into a useful format for analysis. It is also called Data Munging.

Steps of Data Wrangling :-

1) Data Acquisition - collect Data from different sources.

2) Data cleaning - Remove errors, missing values.

3) Data Exploration - Understand structure and content.

4) Data Transformation - Convert into suitable format

5) Data Loading - Load into analysis tools.

Key operations :-

- clean
- Transform
- Merge
- Reshape.

Importance :-

- Improves data quality and accuracy
- Makes data ready for analysis
- Helps in better decision-making
- saves time in analysis.
- Handles missing and inconsistent data.

2) Explain String Manipulation in python with suitable code examples.

String Manipulation refers to processing and modifying text Data using python functions and methods.

Basic String Methods.

```
Val = 'a,b,c'
# split string
Val.split(',')
# ['a','b','c']
# Remove spaces
pieces = [x.strip() for x in Val.split(',')]
# Join strings
':'.join(pieces)
```

2) Searching Operations.

```
Val = 'a, b, c'
'c' in Val        # True
Val.find(',')     # 1
Val.count(',')    # 2
```

3) Replace operation

```
Val.replace(',', '::')
Val.replace(',', '')
```

Regular Expressions

```
import re
text = "foo bar\t baz"
re.split('\s+', text)
# ['foo', 'bar', 'baz']
```

Pandas vectorized string functions.

```
import pandas as pd
import numpy as np
data = pd.Series({
        'Dave': 'dave@google.com'
        'Steve': 'steve@google.com'
        'Rob': 'Rob@google.com'
        'Wes': 'wes@google.com'
    })
data.str.contains('google')
```

3) Explain the general techniques and programming practices used for efficient handling of large data sets in data science

→ Handling large data sets involves efficient techniques and programming practices to manage performance & resources.

Problem with large data:

→ data quality issues
→ security & privacy concerns
→ High cost
→ data integrity issues
→ difficulty in analysis.

Techniques for handling

1) choose the right algorithm.
2) choose the right Data Structure
3) choose the Right Tools

General programming Tips

→ Do not reinvent the wheel
→ Get maximum use of hardware
→ Reduce computational needs.

4) Explain different Data transformation techniques used in preprocessing

→ Data transformation is the process of converting data into a suitable format for analysis.

Techniques:

1) Removing Duplicates

df.drop_duplicates()

2) Transforming Data (Mapping)

df['col'] = df['col'].map(lambda x: x*2)

3) Replacing values

df.replace(-aaa, 0)

4) Renaming Index / Columns

df.rename(columns={'old', 'new'})

5) Binning (Discretization)

pd.cut(data, bins=[0, 50, 200])

6) Standardization

df['z'] = (df['x'] - df['x'].mean()) / df['x'].std()

7) Outlier Detection

Using IQR or Z-Score

8) Creating Dummy Variables

pd.get_dummies(df['col'])